Sahaj Bhadja

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Summary

Early-career Software Engineer with three internships building **production systems**, **REST APIs**, and scalable applications using **Python**, **Java**, SQL, AWS, Docker, microservices, automated testing, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Architected a multi-tenant platform using Azure SQL, Blob Storage, row-level security, and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization.
- Restructured fault-tolerant **asynchronous workflows** using FastAPI workers, processing 1K+ jobs with explicit state transitions and failure isolation while increasing throughput threefold under concurrent production workloads without cascading failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Implemented a production **AWS backend** serving 150+ instructors and 10K+ daily requests, using automated deployments and rubric workflows to reduce feedback turnaround 60% while maintaining stable peak-load performance.
- Designed concurrent **REST APIs** with Python multithreading, AWS RDS, and secure file ingestion, processing 1K+ files per batch through fault-tolerant execution, validation, and reliable customer-facing grading workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Developed a MERN platform with WebSockets, REST APIs, and eight **microservices**, synchronizing 2M+ records across authentication, billing, permissions, and notification workflows while reducing real-time data conflicts 33%.
- Automated Docker packaging and Jenkins **CI/CD** deployments, cutting release time 50%, while Azure SQL and analytics insights reduced resource waste 42% and increased customer lead conversion 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Engineered a cross-platform data-pipeline framework using Databricks, Flask APIs, and Firebase, enabling real-time analytics across 50K+ records with automated testing, cloud synchronization, and integrated Web3 authentication support.
- Designed system architecture and end-to-end workflows, mapping 12+ backend integration points and presenting technical tradeoffs to client stakeholders before implementation, improving project alignment and reducing design errors 25%.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++, C#, JSON
Backend & Data: REST APIs, FastAPI, Flask, Node.js, Spring Boot, OOP, Data Structures, Algorithms, PostgreSQL, MySQL, SQL Server, MongoDB
Cloud & Systems: AWS, Azure, GCP, Docker, Kubernetes, Microservices, Distributed Systems, Data Modeling, Scalability, Performance Optimization
Engineering: Unit Testing, Integration Testing, PyTest, Mockito, Git, GitHub Actions, Jenkins, CI/CD, Monitoring, Code Reviews, Agile/Scrum, Generative AI

Projects

DashMatch Engine | Python, Algorithms, Real-Time Systems

- Built and tested an **object-oriented dispatch engine** using priority queues for nearest-courier assignment, capacity constraints, idempotent delivery handling, latency metrics, and explicit no-capacity failure paths across marketplace scenarios.

Chat2SQL Code | Java, TypeScript, SQL, Docker

- Worked across a **Java** and TypeScript codebase with Docker, Maven, Node.js, and Yarn, supporting 10+ relational and NoSQL databases through schema exploration, query authoring, and cross-platform workflows.

Linguistix | Python, AWS, Docker, CI/CD

- Developed and deployed a containerized RESTful data-processing API on AWS ECS/Fargate with CodePipeline, automated CI/CD releases, validation checks, 87% accuracy, and production-grade reliability across repeatable cloud deployments.